EE 470 Course Project

Spring

Plant Disease Classification Challenge

Human society needs to increase food production by an estimated 70% by 2050 to feed an expected population size that is predicted to be over 9 billion people. Currently, infectious diseases reduce the potential yield by an average of 40% with many farmers in the developing world experiencing yield losses as high as 100%. The widespread distribution of smartphones among crop growers around the world with an expected 5 billion smartphones by 2020 offers the potential of turning the smartphone into a valuable tool for diverse communities growing food. One potential application is the development of mobile disease diagnostics through machine learning and crowdsourcing. In this project, you will use over 50,000 expertly curated images of healthy and infected leaves of crop plants through the existing online platform.

In this course project, you will implement a deep learning algorithm for the problem of plant leaf disease identification. Your algorithms will be evaluated on unseen leaf samples using an accuracy metric. You will compete with each other in getting a better classification algorithm for leaf images. You can use the training leaf images where you can evaluate your performance yourself, but the final evaluation will be on another set of leaf images. You must use Python to implement your codes. To classify plant disease from the leaf image, you can use any image classification algorithm from the literature.



Figure 1. Example of leaf images from the dataset.

Dataset:

The shared ***train_images*** folder contains subfolders (https://drive.google.com/file/d/1UOzt9yuE67uAOQMzZ5F9l76_lwqa8Nm_/view?usp=drive_link). Each subfolder under the "train" folder represents different classes and these subfolders contain RGB retail product images belonging to the product classes. All leaf images contain different difficulties (e.g. colored background and shadows).

Grading Procedure:

The most successful team will get the highest point; the rest will be sorted according to the accuracy of their results. The idea of a challenge competition is to compare the results of various algorithms using a common dataset and a common performance evaluation method.

Project groups: Groups are to have at most **two** students. Following is a link to an online form where you can add the names of your group members. https://docs.google.com/spreadsheets/d/1QwWZG7wRo9i1_El2ztc6HBZwuAZbjqF910ILxF64REY/edit?usp=sharing

EE468 Project Timeline:

Final Project Due: (21st May)

Finalize your project and present it in class. Submit your final report and codes by the same date.